Please read the instructions below before the exam starts, and listen carefully to the instructions given by the invigilators.

You may read the question paper before the exam starts but you must not write anything until the invigilator says that you may start.

You will be given five minutes after the exam has finished to complete the front of any answer books used.

May/June 2024 **EC113 2023/4 A 800**

Only Casio FX-83GTPLUS or FX-85TGPLUS or FX-83GTx or FX-85GTx Casio FX- 85MS calculators are permitted

UNIVERSITY OF READING

Introductory Microeconomics (EC113)

Three hours

Answer **ALL** questions from Section A (32 marks), answer **TWO** questions from Section B (34 marks), and answer **TWO** questions from Section C (34 marks).

Section A

Answer **ALL** questions

1. Consider a consumer who has well behaved preferences for two goods - ice cream cones, c, and bags of popcorn, p. The consumer is consuming where the MRS_{cp} is greater than the price ratio, $\frac{P_c}{P_p}$. Illustrate this with a diagram showing whether the consumer is maximising their utility. If not, what adjustment could be made so that utility is maximised?

(3 marks)

2. Giffen goods have upward sloping demand curves. Assume good x is a Giffen good and y is a normal good. Derive the demand curve for x using two diagrams, one with the axes labelled by quantity demanded for the two goods, x and y. The second showing price and quantity for good x.

(4 marks)

3. A government tax on sugar will reduce consumption of sugary foods and help the obesity problem. What are the positive and normative components of this statement?

(3 marks)

4. Explain the meaning of elasticity of demand. Give a mathematical expression for elasticity of demand. What value would ϵ_d take if it were elastic, and what value would it take if it were inelastic. Comment on the sign.

(3 marks)

5. In the context of consumer demand, consider two normal goods X and Y. Explain what is meant by 'normal' in this context. Describe the effect on demand, in terms of income and substitution effects if the price of good X rises and the price of good Y remains the same. Use a labelled diagram to illustrate your answer.

(3 marks)

6. Explain the term “marginal rate of technical substitution.” What does a MRTS = 4 mean?

(3 marks)

7. Joe quits his computer programming job, where he was earning a salary of £50,000 per year, to start his own computer software business in a building that he owns and was previously renting out for £24,000 per year. In his first year of business he has the following expenses: salary paid to himself, £40,000; rent, £0; other expenses, £25,000. Find the accounting cost and the economic cost associated with Joe’s computer software business.

(3 marks)

8. Several years ago, Alcoa was effectively the sole seller of aluminium because the firm owned nearly all of the aluminium ore reserves in the world. This market was not perfectly competitive. Why? Which assumptions are violated?

(2 marks)

9. Consider the following game in which two firms decide how much of a homogeneous good to produce. The annual profit (payoffs) for each firm are stated in the cell of the game matrix, and Firm A's payoffs appear first in the payoff pairs: What is the Nash equilibrium for this game?

	Firm B - low output	Firm B - high output
Firm A - low output	300, 250	200, 100
Firm A - high output	200, 75	75, 100

(3 marks)

10. Assume that a profit maximizing monopolist is producing a quantity such that marginal revenue exceeds marginal cost. What can we conclude from this? How can this monopolist maximise profit?

(3 marks)

11. When government intervenes in a competitive market by imposing an effective price ceiling, we would expect the quantity supplied to _____ and the quantity demanded to _____.

(2 marks)

Section B

Answer **TWO** questions

12. Mary likes Coke (c) and Pepsi (p). Assume Mary spends all her money, M, on these two goods. Denote the price of Coke as p_c and the price of Pepsi as p_p.

(a) First, assume Mary has well behaved preferences. State the assumptions you need to make for preferences to be well-behaved. Give a brief explanation for each.

(3 marks)

(b) Graphically represent a general utility maximising bundle for Mary with demand for Coke on the horizontal axis and Pepsi on the vertical axis. Be sure to label the diagram properly, including intercepts and slopes.

(4 marks)

(c) Now assume Mary has preferences such that she is indifferent between Coke and Pepsi. Draw a map of Mary's indifference curves with demand for Coke on the horizontal axis and Pepsi on the vertical axis.

(2 marks)

(d) If Mary's preferences are as in part c above, and the price of Coke is greater than the price of Pepsi, what will Mary's utility maximising consumption bundle be? Why? Illustrate the outcome with a diagram. Be sure to fully label the diagram.

(4 marks)

(e) Coke and burgers are compliments. If the price of Coke increases, what would happen to quantity demanded for burgers? Answer this with two diagrams. Give an expression for the cross-price elasticity of Coke and Burgers. Remember to show the sign.

(4 marks)

13. Consider a firm, which produces a good new sort of phone charger, X. The firm faces a supply curve;

$$Q^S = 204 + P$$

In the beginning there is only one buyer for the phone charger. Their market demand Q^D, for good X is given by:

$$Q^D = 240 - 2P$$

(a) Calculate the equilibrium price and quantity. Illustrate this equilibrium in a diagram with P on the vertical axis and Q on the horizontal axis. Include labels for the vertical intercepts, equilibrium price and quantity, and slope of the curves. Remember you will plot the *inverse* demand and supply curves. Show your workings.

(5 marks)

(b) The product X turns out to be popular. Assume new buyers enter the market. If there are now 5 buyers with identical demand curves ($n = 5$) in the market for X, calculate the new market demand curve and the inverse market demand curve.

(4 marks)

(c) Assume that the supply curve is as in part (a) above. Show, on a diagram with P on the vertical axis, and Q on the horizontal axis, the market inverse demand curve when $n = 5$ and the inverse supply curve. Label the vertical intercepts and the slopes of the curves and explain your findings.

(2 marks)

(d) What is the market equilibrium price and quantity now that there are 5 buyers for X? Show your workings and label these equilibrium values on the diagram in part (c).

(3 marks)

(e) Now suppose that more firms begin to *supply* good X. Provide a general expression for the *inverse supply curve* if n firms, with identical supply curves, enter the market. If demand remains the same, what is the general effect for the buyers of X of increasing number of suppliers, in terms of of equilibrium price and quantity?

(3 marks)

14. Chloe consumes only books (x) and video games (y). Her preferences can be represented by the following utility function: $U\ (x, y)\ =\ xy$. The price of books is p_x, the price of video games is p_y, and Chloe has an income of M dollars.

(a) Write down Chloe's budget constraint and budget line, assuming for the budget line that demand for video games is on the vertical axis and demand for books is on the horizontal axis.

(2 marks)

(b) Provide a general expression for the Marginal Rate of Substitution between goods *x* and *y*. Explain the expression and comment on what the marginal rate of substitution tells us about Chloe's preferences and how it is represented graphically.

(2 marks)

(c) What is the condition that must be satisfied for Chloe to be consuming a bundle that maximises her utility? Provide a general expression for this.

(2 marks)

(d) If $MU_x = y$ and $MU_y = x$, give an exact expression for the MRS_{xy} from $U(x, y)\ =\ xy$. If Chloe is consuming 20 units of y and 4 units of x, and the price of x is £1.50, what price would y have to take for Chloe to be maximising her utility at this bundle?

(4 marks)

(e) Suppose the price of books increases. Assuming both goods are normal goods, show, with the aid of a diagram, how Chloe's utility maximising bundle changes. Decompose the effect of the price change into the income effect and the substitution effect and label these on your diagram.

(4 marks)

(f) What is an inferior good? Give two examples. Show, on a separate diagram to the one in part (e), how the utility maximising bundle changes when the price of books falls, if books are an inferior good. Comment on the outcome.

(3 marks)

Section C

Answer **TWO** questions

15. Dishi Metal Company produces aluminium fittings. Dishi's engineers estimate the production function represented below as relevant for their long-run capital-labour decisions:

$Q = 500L^{0.5}K^{0.5}$

where Q = annual output measured in pounds,

L = labour measured in person hours,

K = capital measured in machine hours.

Dishi's employees are relatively highly skilled and earn £15 per hour. The firm estimates a rental charge of £50 per hour on capital. Dishi forecasts annual costs of £500,000 per year, measured in real pounds.

(a) What are the marginal products of labour and capital?

(4 marks)

(b) Determine the firm's optimal capital-labour ratio, given the information above.

(4 marks)

(c) How much capital and labour should the firm employ, given the £500,000 budget? Calculate the firm's output.

(5 marks)

(d) Dishi is currently negotiating with a newly organised union. The firm's personnel manager indicates that the wage may rise to £25 under the proposed union contract. Analyse the effect of the higher union wage on the optimal capital-labour ratio and the firm's employment of capital and labour. What will happen to the firm's output?

(4 marks)

16. This question examines the economics of a minimum wage and wage subsidies. Suppose the supply of low-skilled labour is given by $L^S = 10w$, where L^S is the quantity of low-skilled labor (in millions of persons employed each year), and w is the wage rate (in pounds per hour). The demand for labour is given by $L^D = 80 - 10w$.

(a) What will be the free-market wage rate and employment level? Suppose the government sets a minimum wage of £5 per hour. How many people would then be employed?

(7 marks)

(b) Suppose that instead of a minimum wage, the government pays a subsidy of £1 per hour for each employee. What will the total level of employment be now? What will the equilibrium wage rate be?

(10 marks)

17. The widget market is controlled by two firms: Acme Widget Company and Widgetway Manufacturing. The structure of the

market makes secret price cutting impossible. Each firm announces a price at the beginning of the time period and sells widgets at the price for the duration of the period. There is very little brand loyalty among widget buyers so that each firm's demand is highly elastic. Each firm's prices are thus very sensitive to inter-firm price differentials. The two firms must choose between a high and low price strategy for the coming period. Profits (measured in thousands of dollars) for the two firms under each price strategy are given in the payoff matrix below. Widgetway's profit is before the comma, Acme's is after the comma.

		Acme	
		Low Price	High Price
Widgetway	Low Price	60, 60	250, -20
	High Price	-20, 250	130, 130

(a) Does either firm have a dominant strategy? What strategy should each firm follow?

(5 marks)

(b) Assume that the game is to be played an infinite number of times. (Or, equivalently, imagine that neither firm knows for certain when rounds of the game will end, so there is always a positive chance that another round is to be played after the present one.) Would the tit-for-tat strategy be a reasonable choice? Explain this strategy.

(6 marks)

(c) Assume that the game is to be played a very large (but finite) number of times. What is the appropriate strategy if both firms are always rational?

(6 marks)

(End of question paper)